

## GRADISTE, Croatia

WMO: 143820

Lat: 45.1592N

Lon: 18.7036E

Elev: 98

StdP: 100.15

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -10.4      | -7.5 | -13.4                         | 1.2 | -9.1 | -10.7 | 1.5 | -6.3 | 8.4                   | 3.3  | 7.4 | 2.5  | 1.9                   | 70   | 0.431 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.9                   | 34.3            | 21.8 | 32.6 | 21.5 | 30.9 | 21.0 | 23.4                | 30.9 | 22.5 | 30.1 | 21.8 | 29.1 | 2.2                  | 100  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.0                            | 15.9 | 26.6 | 20.2 | 15.0 | 25.9 | 19.3 | 14.3 | 25.0 | 70.0          | 31.2 | 66.7 | 30.2 | 63.9 | 29.4 | 26.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 7.4               | 6.2  | 5.2 | -14.1                      | 36.4 | 4.6                | 2.3 | -17.4                                              | 38.1 | -20.1      | 39.4 | -22.7      | 40.7 | -26.0      | 42.4 |  |  |
|                   |      |     | -14.5                      | 24.9 | 4.5                | 0.8 | -17.7                                              | 25.5 | -20.4      | 25.9 | -22.9      | 26.4 | -26.2      | 26.9 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 12.6   | 1.4   | 2.8   | 7.8   | 13.2  | 17.3  | 21.2  | 22.8  | 22.7  | 17.8  | 12.8  | 8.0   | 2.6  |
|                                                                                  | DBStd       | 8.68   | 5.09  | 5.30  | 4.87  | 3.76  | 3.74  | 3.60  | 3.14  | 3.13  | 3.75  | 4.17  | 4.49  | 4.41 |
|                                                                                  | HDD10.0     | 932    | 269   | 204   | 100   | 14    | 2     | 0     | 0     | 0     | 0     | 21    | 89    | 231  |
|                                                                                  | HDD18.3     | 2558   | 526   | 434   | 328   | 158   | 64    | 15    | 3     | 3     | 53    | 176   | 310   | 486  |
|                                                                                  | CDD10.0     | 1877   | 2     | 3     | 31    | 110   | 227   | 335   | 398   | 395   | 235   | 108   | 30    | 3    |
|                                                                                  | CDD18.3     | 460    | 0     | 0     | 0     | 3     | 31    | 99    | 143   | 140   | 38    | 4     | 0     | 0    |
|                                                                                  | CDH23.3     | 4798   | 0     | 0     | 2     | 66    | 331   | 994   | 1491  | 1484  | 382   | 48    | 1     | 0    |
|                                                                                  | CDH26.7     | 1898   | 0     | 0     | 0     | 11    | 81    | 369   | 642   | 673   | 119   | 3     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.2    | 2.2   | 2.5   | 2.7   | 2.4   | 2.4   | 2.1   | 2.1   | 1.9   | 2.0   | 1.9   | 2.0   | 2.2  |
| Precipitation                                                                    | PrecAvg     | 745    | 48    | 45    | 48    | 62    | 77    | 83    | 75    | 63    | 68    | 59    | 62    | 57   |
|                                                                                  | PrecMax     | 1004   | 88    | 82    | 85    | 163   | 178   | 235   | 212   | 184   | 206   | 147   | 116   | 138  |
|                                                                                  | PrecMin     | 364    | 10    | 6     | 7     | 11    | 28    | 34    | 14    | 2     | 0     | 0     | 6     | 2    |
|                                                                                  | PrecStd     | 172    | 21    | 20    | 23    | 36    | 41    | 45    | 47    | 45    | 48    | 40    | 30    | 33   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 15.3  | 18.1  | 22.9  | 28.5  | 31.3  | 34.2  | 36.6  | 37.2  | 32.8  | 27.0  | 21.9  | 16.3 |
|                                                                                  |             | MCWB   | 10.6  | 10.8  | 12.8  | 16.8  | 19.1  | 22.5  | 22.0  | 21.6  | 20.0  | 17.5  | 14.9  | 11.6 |
|                                                                                  | 2%          | DB     | 12.4  | 14.6  | 20.6  | 25.1  | 28.9  | 32.4  | 34.2  | 34.6  | 29.9  | 24.6  | 19.4  | 13.6 |
|                                                                                  |             | MCWB   | 8.7   | 9.4   | 11.9  | 14.8  | 18.9  | 22.0  | 22.0  | 21.6  | 19.0  | 16.7  | 13.7  | 9.9  |
|                                                                                  | 5%          | DB     | 10.1  | 12.3  | 18.3  | 22.9  | 27.0  | 30.6  | 32.4  | 32.5  | 27.3  | 22.4  | 16.8  | 10.9 |
|                                                                                  |             | MCWB   | 7.5   | 8.2   | 11.0  | 14.0  | 18.1  | 21.3  | 22.0  | 21.2  | 18.3  | 15.9  | 12.5  | 8.2  |
|                                                                                  | 10%         | DB     | 7.8   | 9.8   | 15.8  | 20.8  | 24.9  | 28.8  | 30.4  | 30.6  | 25.2  | 20.0  | 14.6  | 8.5  |
|                                                                                  |             | MCWB   | 5.8   | 7.0   | 9.8   | 13.0  | 16.9  | 20.4  | 21.2  | 20.6  | 17.7  | 14.8  | 11.5  | 6.6  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 10.7  | 11.3  | 13.7  | 17.2  | 20.7  | 24.1  | 24.4  | 24.2  | 21.2  | 18.7  | 15.4  | 12.1 |
|                                                                                  |             | MCDB   | 14.8  | 16.6  | 21.0  | 27.1  | 27.7  | 31.0  | 31.9  | 31.1  | 29.5  | 25.0  | 20.7  | 15.6 |
|                                                                                  | 2%          | WB     | 9.1   | 9.9   | 12.5  | 15.7  | 19.7  | 22.9  | 23.5  | 23.1  | 20.1  | 17.4  | 14.1  | 10.1 |
|                                                                                  |             | MCDB   | 12.0  | 14.1  | 18.6  | 23.3  | 26.6  | 30.0  | 31.1  | 31.1  | 27.4  | 22.9  | 17.9  | 13.0 |
|                                                                                  | 5%          | WB     | 7.5   | 8.5   | 11.5  | 14.7  | 18.7  | 22.1  | 22.6  | 22.2  | 19.2  | 16.3  | 13.0  | 8.4  |
|                                                                                  |             | MCDB   | 9.7   | 11.5  | 17.1  | 21.3  | 25.5  | 29.0  | 30.2  | 30.0  | 25.3  | 21.2  | 16.7  | 10.7 |
|                                                                                  | 10%         | WB     | 6.0   | 7.2   | 10.3  | 13.7  | 17.7  | 21.2  | 21.8  | 21.4  | 18.2  | 15.4  | 11.6  | 6.8  |
|                                                                                  |             | MCDB   | 7.6   | 9.5   | 15.4  | 19.4  | 23.9  | 27.5  | 29.2  | 28.6  | 23.7  | 19.4  | 14.4  | 8.4  |
| Mean Daily<br>Temperature<br>Range                                               |             | MDBR   | 6.1   | 7.2   | 9.5   | 11.0  | 11.0  | 11.0  | 11.9  | 12.5  | 11.1  | 10.3  | 8.0   | 5.5  |
|                                                                                  | 5% DB       | MCDBR  | 9.8   | 11.6  | 14.1  | 15.4  | 14.1  | 13.8  | 15.1  | 16.2  | 15.0  | 14.2  | 11.6  | 9.6  |
|                                                                                  |             | MCWBR  | 6.7   | 7.2   | 7.3   | 7.0   | 5.8   | 5.2   | 5.2   | 5.6   | 5.8   | 7.1   | 6.9   | 6.5  |
|                                                                                  |             | MCDBR  | 9.4   | 10.5  | 12.5  | 13.3  | 13.0  | 12.8  | 13.6  | 14.0  | 12.9  | 13.0  | 10.7  | 9.0  |
|                                                                                  | 5% WB       | MCWBR  | 6.6   | 6.7   | 6.9   | 6.3   | 5.7   | 5.2   | 5.2   | 5.5   | 5.3   | 6.8   | 6.5   | 6.4  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.341  | 0.369 | 0.401 | 0.458 | 0.445 | 0.442 | 0.445 | 0.448 | 0.431 | 0.397 | 0.360 | 0.341 |      |
|                                                                                  | taud        | 2.356  | 2.272 | 2.221 | 2.073 | 2.169 | 2.218 | 2.206 | 2.203 | 2.217 | 2.307 | 2.364 | 2.363 |      |
|                                                                                  | Ebn at Noon | 753    | 797   | 822   | 802   | 828   | 831   | 822   | 804   | 781   | 755   | 722   | 710   |      |
|                                                                                  | Edh at Noon | 78     | 102   | 123   | 154   | 145   | 139   | 139   | 134   | 122   | 98    | 78    | 71    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.25   | 2.02  | 3.27  | 4.50  | 5.51  | 6.07  | 6.17  | 5.51  | 3.82  | 2.63  | 1.54  | 1.01  |      |
|                                                                                  | RadStd      | 0.19   | 0.28  | 0.41  | 0.46  | 0.49  | 0.52  | 0.47  | 0.56  | 0.43  | 0.28  | 0.14  | 0.16  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | +0.85 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (36 neighbors) | +0.67 | N/A     | N/A    | N/A     | +0.41 | +0.43 | -120        | -183    | +115    | +61     |

Nomenclature: See separate page